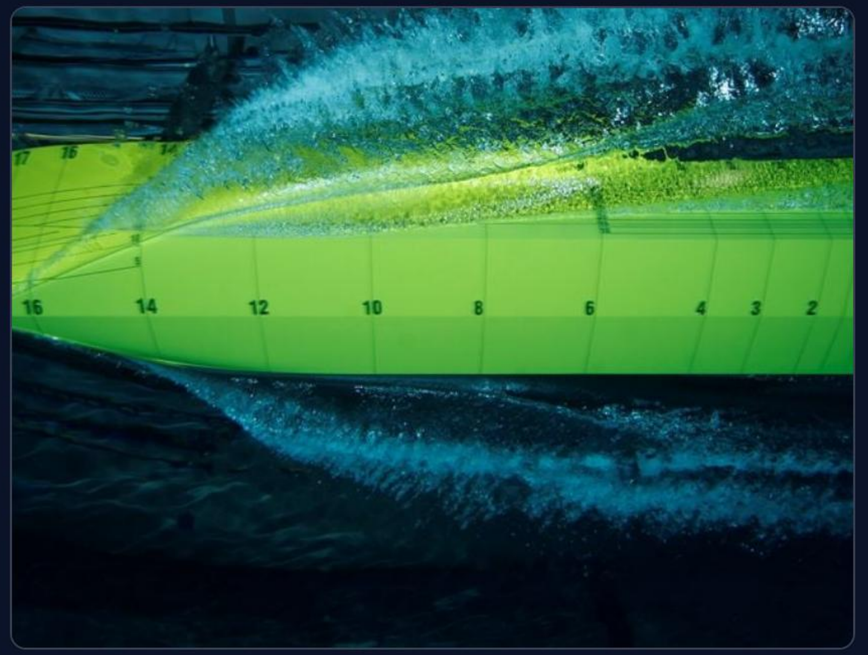
CONTROLLED AI ENGINEERING

NAVIS AI

Naval Assistance & Verification Intelligence System

AI-assisted programming with verification, traceability, and engineer control for early-stage naval research.



WORKFLOW FORMULA | AI Proposes → Evidence Verifies → Engineers Decide

CONCEPTUAL PROPOSAL

EXECUTIVE SUMMARY

The Core Opportunity & Challenge

The Challenge

AI accelerates programming, debugging, and documentation—but risks outputting inaccurate, insecure, or overconfident code.

Core Constraint: AI must never hold autonomous authority.

The Proposal

NAVIS AI acts as a controlled assistant that links evidence, runs verification checks, and routes all outputs for engineer sign-off.

Target Outcome: Faster development without lost oversight.



Governing Principle

"AI is an assistant—not an authority, certifier, or autonomous operator."

DOMAIN CONTEXT

Understanding Carderock's Mission

SCOPE

Naval Engineering Support

Full-spectrum research in naval architecture, hull dynamics, ship integration, and survivability.

TARGET TASKS

Early R&D Workflows

Accelerating code creation, debugging, unit testing, documentation, and technical onboarding.

BOUNDARY

Non-Negotiable Constraints

Preserving security, ethical accountability, and mandatory human authorization on all commits.

Primary User Community

Scientists, naval architects, software engineers, and Science & Technology project managers.

RISK ANALYSIS

Writing Code vs. Trusting Results

Hallucinations

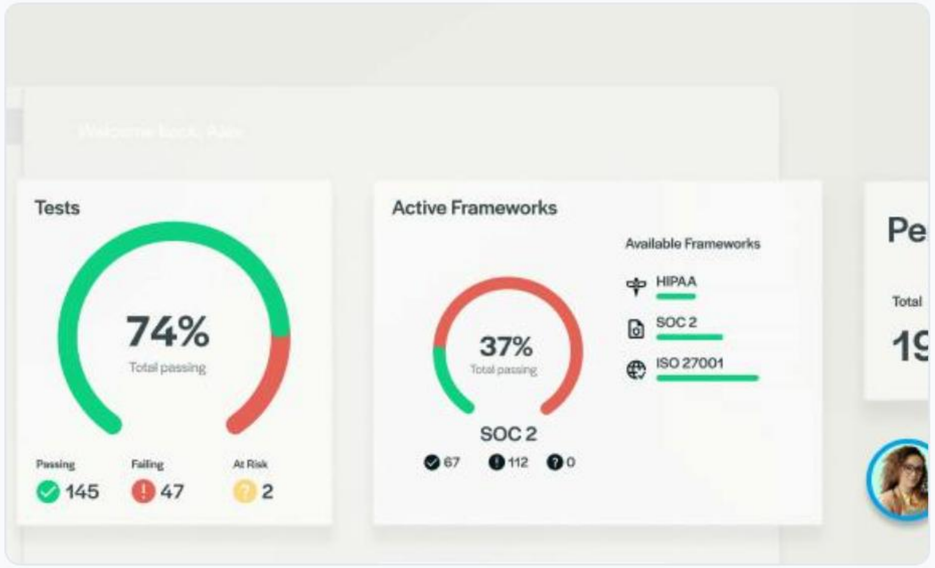
Plausible explanations or code snippets containing subtle mathematical or logic errors.

Security Vulnerabilities

Generated code introduces unvalidated inputs, insecure dependencies, or hidden defects.

Over-Reliance

Engineers accepting assistant outputs without performing rigorous manual code review.



Core Defense Mechanism

Separate AI Generation from Verification via Deterministic Sandboxing

ARCHITECTURE

NAVIS AI: 5-Stage Controlled Workflow

01

Intake

User selects task and provides approved, non-sensitive inputs.

02

Analyze

AI identifies task type, checks assumptions, and flags risk levels.

03

Propose

AI drafts code solution with rationale and evidence links.

04

Verify

Sandbox runs static analysis, unit tests, and security scans.

05

Approve

Engineer reviews evidence and accepts or rejects output.

No Automatic Commits

Mandatory Human Review

Complete Audit Trail

DEMONSTRATION SCENARIO

Sample Workflow: Debugging Code

INPUT & PROPOSAL

User Input

"Find why sample Python function returns incorrect average on synthetic dataset."

AI Proposal

"Division by zero occurs on empty list inputs. Proposed fix: add explicit boundary validation."

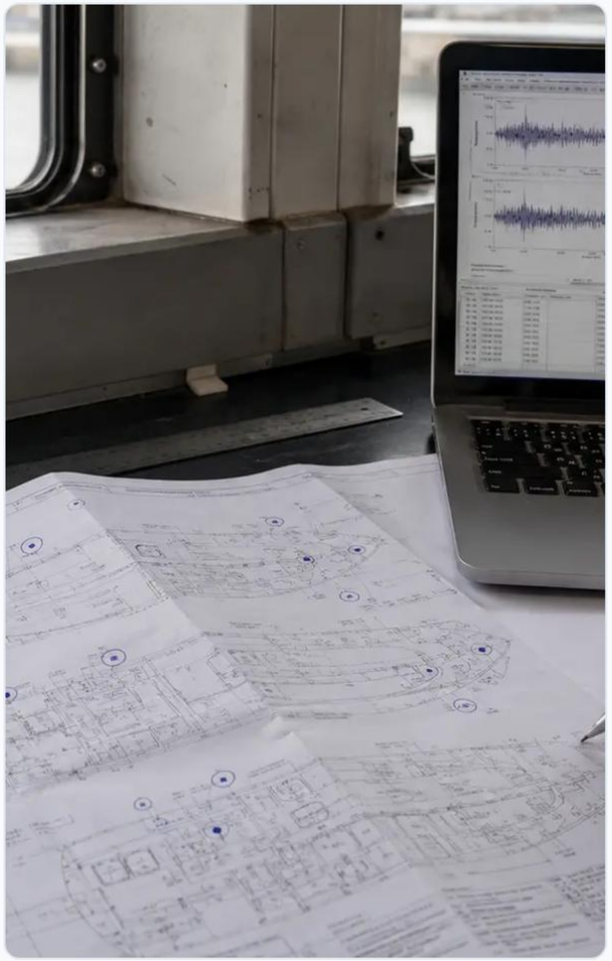
Synthetic non-sensitive demonstration case.

AUTOMATED CHECKS

Verification Execution

- Automated unit tests (empty, single, multiple values)
- Static analysis & security linting passed
- Engineer reviews proposed patch & test log

Result: Accepted by Engineer with full log trace.



Demonstration scenario only — does not represent active deployment on Navy systems.

GOVERNANCE CONTROLS

Built-In System Safeguards

Data Protection

Restricted to approved environments. Classified or sensitive data automatically blocked.

Fail-Closed Reliability

Independent security tools scanning. System pauses if tests cannot run.

Full Traceability

Log prompts, model version, verification evidence, and reviewer approval.

Human Authority

Zero autonomous commits or system changes without explicit engineer approval.

Policy Rule: "If the system cannot establish that a task is safe and in scope, it must pause and request human review."

DECISION LOGIC

Risk-Based Request Routing

TIER 1 • LOW RISK

Learning & Reference

Language syntax explanations or simple code examples.

Action: Standard response with manual user check.

TIER 2 • MODERATE RISK

Engineering Support

Debugging research code, drafting unit tests.

Action: Mandatory sandbox tests & engineer review.

TIER 3 • RESTRICTED

Operational & High Impact

Classified data, safety-critical operational code.

Action: Blocked / Escalated. No AI action.

Note: Risk classification matrix subject to final Carderock governance policy.

IMPLEMENTATION STRATEGY

Practical Phased Pilot Roadmap

PHASE 1

Discover

Select low-risk use cases; align data rules with cybersecurity & legal teams.

PHASE 2

Prototype

Deploy sandbox demo with synthetic code, routing, and audit logs.

PHASE 3

Evaluate

Benchmark correctness, security findings, time savings, and override rates.

PHASE 4

Governed Expansion

Expand only upon formal security authorization and documented performance.

Controlled expansion protects existing software assurance standards.

PERFORMANCE MEASUREMENT

Evaluation Criteria Framework

Metric Category	Primary Target Indicator
Correctness	Percentage of suggestions passing automated tests
Security Quality	Vulnerabilities introduced, missed, or flagged
Task Efficiency	Time saved compared against manual baseline
Human Oversight	Review completion rate & rejection frequency
Traceability	Percent of outputs with complete evidence records

Baseline metrics must be established prior to pilot rollout; performance gains will not be assumed.

STRATEGIC SUMMARY

The Core Recommendation

Adopt a Controlled Pilot — Not Unchecked AI

NAVIS AI enables Carderock to explore productivity gains in programming while enforcing security, verification, and human oversight required for naval systems.

AI proposes. Evidence verifies. Engineers decide.

DOCUMENTATION

Selected References & Standards

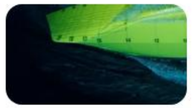
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DON CIO Generative AI Guidance (2023)
Department of the Navy Chief Information Officer LLM policy.

NIST AI Risk Management Framework (2024)
Generative AI Profile (NIST AI 600-1) for system verification.

IEEE & ACM Software Security Studies (2024-2025)
Empirical research on defect rates & security practices in AI coding.

Image Sources



https://www.sintef.no/contentassets/4d6d574ace8141ba95d88a662b6fb8f9/dwl-45kn_1trim.jpg?width=1600&height=500&mode=crop&quality=80

Source: www.sintef.no



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